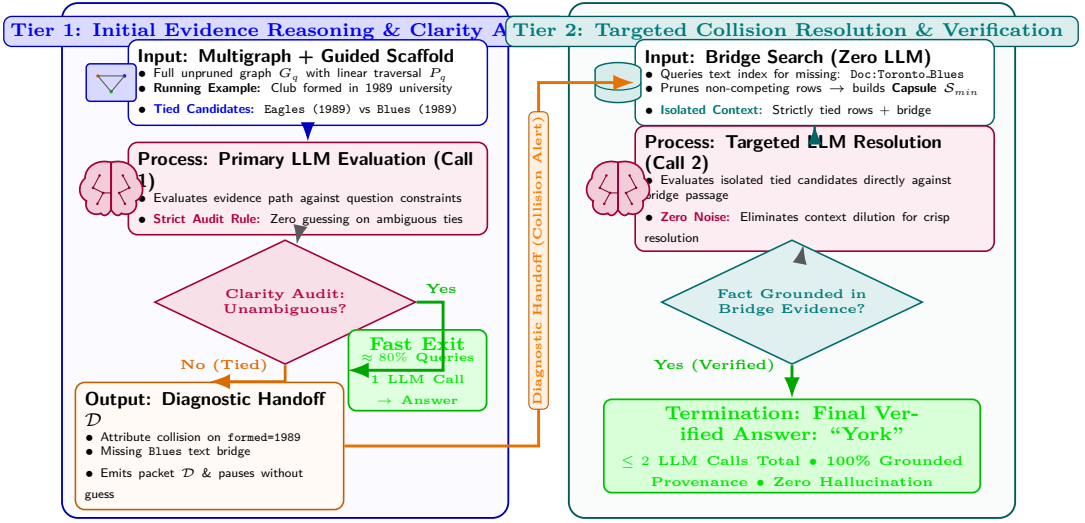

Tier 1: Initial Evidence Reasoning & Clarity A
Input: Multigraph + Guided Scaffold
Full unpruned graph G_q with linear traversal P_q
Running Example: Club formed in 1989 university
Tied Candidates: Eagles (1989) vs Blues (1989)
Process: Primary LLM Evaluation (Call 1)
Evaluates evidence path against question constraints
Strict Audit Rule: Zero guessing on ambiguous ties
Clarity Audit: Unambiguous?
Yes
Fast Exit
≈ 80% Queries
1 LLM Call
→ Answer
No (Tied)
Output: Diagnostic Handoff $\mathcal{D}$
Attribute collision on formed=1989
Missing Blues text bridge
Emits packet $\mathcal{D}$ & pauses without guess
Diagnostic Handoff (Collision Alert)
Tier 2: Targeted Collision Resolution & Verification
Input: Bridge Search (Zero LLM)
Queries text index for missing: Doc:Toronto_Blues
Prunes non-competing rows → builds Capsule $\mathcal{S}_{min}$
Isolated Context: Strictly tied rows + bridge
Process: Targeted LLM Resolution (Call 2)
Evaluates isolated tied candidates directly against bridge passage
Zero Noise: Eliminates context dilution for crisp resolution
Fact Grounded in Bridge Evidence?
Yes (Verified)
Termination: Final Verified Answer: "York"
≤ 2 LLM Calls Total • 100% Grounded Provenance • Zero Hallucination